This document is intended to provide a basis for Midterm Senior Project Evaluations and is intended for use by the Faculty Coach, the Project Team, and any stakeholders, including the Project Sponsor. The evaluation criteria below are divided into two main areas: Process/Project, and Product. Early in the project, the primary focus was placed on the Process/Project areas; in this evaluation we will focus much more heavily on product. Because each area is expansive and often includes entire classes covering them, a sample rubric, some questions, and general evaluation categories have been provided to increase understanding of what is expected throughout the project.

To use this document, go through each area and facilitate discussion. It is not necessary to complete each rubric or to fill in each question, but students should be encouraged to individually do so, meet as a group for comparison, and report back to the faculty coach and sponsor any results or conclusions they reach.

Process and Project

The tendency to focus on the product often leaves the process and the project out in the cold, especially in projects with set or compressed schedules. A well developed Software Engineering team knows that the returns paid within the Process and Project domains far outweigh the time and expenses that occur. Although the list below is not exhaustive, it includes some of the primary areas of importance.

Meetings and Communications Management

Sample Rubric: Meetings and Communications Management				
Meetings were held only at regular intervals or as a result of a crisis and did not include agendas or pre-planning	Meetings included standing agendas which were occasionally followed	Meetings included predetermined agendas, but the agenda was not used within the meeting or evaluated afterward. Included some items which did not necessitate a meeting	Meetings included predetermined agendas and were somewhat organized, but were occasionally allowed to stray from format. Content necessitated a meeting	Meetings were pre organized, included prepared agendas, followed expected format, and did not include content that did not necessitate a meeting

Communication with stakeholders was reactive or crisis driven, did not occur when expected, and did not clearly communicate expectations	Communication was conducted out of necessity but was rarely proactive and did not account for stakeholder expectation management	Communication was scheduled and typically took place. Expectation management was rarely contemplated	Communications with stakeholders were regular and generally proactive. Expectation management was occasionally used.	Communications with stakeholders were regular, proactive, and purpose-driven. Language was clear and included expectation management
In terms of customer education, team had a negative attitude towards the customer and did not bother to assist them	Team blindly accepted customer inputs, did not convey own knowledge adequately	Team missed some occasions when customers needed education or clarifications were needed	Team provided technical inputs to customer when needed or asked for	Team made sure at every stage to educate customer and provide/seek clarifications
Comments:				

1. Is the team conducting effective meetings? What can be changed to make the team meetings more productive?

   The team is conducting highly effective meetings, as these meetings are structured to give the team an overview of the project from leadership's perspective, get all the member's feedback/point's of information they think the rest of the team should know, then get into the content of the meeting. This usually includes reviewing pull requests, designs, wireframes, and architectural overviews. At this point in the project, team meetings could only get more productive if we spent time on peer programming, but the team doesn't believe that's an efficient usage of meeting time and would rather have shorter, effective meetings. Another potential area of improvement would be having more consistent meeting end times. Some meetings can go longer than others, and it isn't made exceptionally clear how long a meeting will take before the meeting itself, so planning and communicating a timeframe is something the team can do more going forward.

2. How well has the team been communicating project progress to the sponsor? What regular communication does the team have with the sponsor? Has the team been maintaining this communication to the satisfaction of the sponsor? Were any adjustments needed in the communication over time? Were these changes initiated by the team or the sponsor?

   The team communicates progress at regular two week intervals with user acceptance testing and a demo of the work. Additionally, the team uses github projects to have a transparent backlog and roadmap for the sponsor to understand the work that's been done and planned. Completed backlog items are moved into a ready for sponsor review section to make it clear what has been worked on. This allows the sponsor to understand the project and what phase all sprint items are in. This communication has been effective in bringing the sponsor up to speed after a sprint, and has been satisfactory. Since the process refinements that were done in the beginning of the summer, no adjustments have been needed.

Sponsor/Stakeholder Communications

Comments: Sponsor agrees with comments, team and coach have no further input.

Time and Effort

Sample Rubric: Time and Effort				
Student effort was minimal, not meeting minimum requirements and putting the success of the project in crisis	Students effort only met minimum standards, and excuses or blame was often used to explain the effect of internal or external hampering factors	Student effort was adequate to meet project requirements, but was frequently hampered by both internal and external factors	Student effort was consistently better than average, even with negative internal factors. External factors occasionally affected effort levels	Students consistently put in full effort, even when internal or external factors have hampered progress
Comments:				

1. Has the level of effort of individual team members been reflected in the output of the team as a whole?

   The individual team members have completed extremely competitive amounts of work each sprint, and by all metrics only improve each sprint. This is reflected in the output, as the product is nearing MVP completion and looks and functions like a professional grade product.

2. To what extent has thrashing (unproductive work) become a factor?

   Thrashing has been a minor factor, especially with early attempts at process improvements. After the process was well defined and agreed on, thrashing was kept to a minimum.

Conformance to Schedule

Sample Rubric: Conformance to Schedule				
Did not demonstrate any schedule management and did not meet schedule requirements	Met minimal schedule requirements but showed no schedule management	Met most schedule requirements with little or no active schedule management	Met most or all schedule requirements with some active schedule management	Actively managed and met all schedule requirements
Comments:				

1. Has the team met all project milestones to date? Which milestones, if any, were missed or were met ahead of schedule? What contributed to this schedule changes? What will the team do differently to ensure that future milestones are met?

   All milestones achieved ahead of schedule, and enhancements have also started being completed. MVP completion is scheduled for 6/30/26.

Change and Scope Management

Sample Rubric: Change and Scope Management				
Did not demonstrate awareness of issues as change or scope issues but rather as individual problems	Occasionally identified change or scope issues as such but often addressed them reactively as individual problems	Addressed change and scope when issues were presented with little or no root cause analysis	Addressed change and scope regularly but not always actively addressing change and scope issues before they occurred	Actively managed change and scope, seeking out and addressing issues before they affected the project. Used root cause analysis
Comments:				

1. Has the scope of the project been considered at each milestone?

   As each milestone was completed, the scope became more ambitious. This was especially so after the summer break, when the team was presented with a proposal involving massive changes in scope. A significant amount of meeting time was invested into analyzing the feasibility of handling these change requests. To address these numerous scope adjustments, the team established a formal management process. This included a primary consensus with the project sponsor acknowledging that while certain modifications may remain unfinished, the group remains dedicated to fulfilling its commitments and establishing a robust technical foundation for any remaining items.

Risk Management

Comments: Clarified that scope change and requirements elaboration requests have been ack. by the team and the team is committed to delivering an updated version of the MVP that ensures legacy devices communicate with the V2 server.

Deliverables

Sample Rubric: Deliverables				
Deliverable was not present and not substituted with other efforts or deliverables	Deliverable was delivered with minimal effort and did not contribute significantly to the success of the project. Some obvious problems or inconsistencies were present, demonstrating little QA occurred	Deliverable met standards and was useful at introduction, but was not updated or useful further along in the project. Few obvious problems or inconsistencies were present	Deliverable was better than average and was used or enhanced irregularly, performing its desired function within the project. No obvious problems or inconsistencies were present	Deliverable performed as expected, was reviewed and enhanced at regular intervals, and would have consistently allowed someone not familiar with the project to become informed
Comments:				

1. How often do these artifacts get updated on the department project website?

   Artifacts get updated on the team's project website (not the department website) whenever a change to major documentation occurs, on a rolling basis. This ensures the website contains all the newest relevant information.

2. Has the team had any issues with configuration management? How were these problems solved? What percentage of project artifacts is under configuration control?

Configuration management has been a topic of major concern, especially as it pertains to security updates for older devices. These problems were solved by deep research into potential patches, as well as weighing different trade offs. Additionally the team found that proposals for documents involving scope change requests - documents whose usage was being explored by the team - had to be partially abandoned. The goal of these documents was to formalize an informal process of change introduction, justification, impact and feasibility assessment, and approval. These documents ended up being implemented internally and have proved effective thus far.

Project Website

Comments:

Risk Management Table

Comments: Coach is good as long as they are effective internally.

Quality

Sample Rubric: Quality				
Little attention was paid to initial requirements and the final product related only loosely to them	Requirements were created and used as a guideline for initial efforts, but were rarely addressed later in the project	Requirements management and quality control were performed irregularly, but there was an understanding of the link between quality and conformance to requirements	Conformance to functional requirements was a priority, but non functional requirements often took a back seat, relating to quality control	Conformance to functional and non functional requirements was clearly a priority and actively prevented scope creep and similar issues

A customer-view of quality was not considered throughout the project	Customer satisfaction was implied as a requirement but not effectively understood by the team. Treated as a requirement rather than a separate quality factor	The group understood that customer satisfaction was important, and occasionally addressed it, but focused primarily on requirements	Customer satisfaction was a priority, and some attention was placed to areas such as maintainability, but some imbalance occurred	Customer satisfaction and a long-term scope as a priority was demonstrated in each project step and properly balanced with conformance to requirements and scope

Comments:

1. Has the work accomplished to date satisfied the project sponsor? Team's opinion? Sponsor's opinion? Were there any weak spots in this regard?

According to sponsor feedback, the work accomplished to date (6/22/2026) has satisfied the sponsor. The team's opinion is that the SWEN MVP will be complete as of the end of the sprint. The only "weak spot" in this regard was the necessity to implement breaking changes to the API to ensure secure authentication, though this was considered important enough to do.

Sponsor: More than excellent! No concerns on quality. Sponsor good at finding bugs, hasn't found many.

Process Creation and Improvement

Sample Rubric: Process Creation and Improvement				
Process is only used when clear best practice standards exist or an external process must be followed	Processes have been created and are usually followed, but are rarely improved.	Processes have been created and are consistently followed, but are rarely improved. Crisis situations occasionally trigger	Processes are always followed and occasionally improved. Root cause analysis may be a byproduct but is not actively pursued	Active process improvement is addressed regularly and processes are evaluated/root cause analysis is used regularly

		process improvement		
Metrics were not planned or used throughout the project	Metrics were used only as required by faculty coach or project guidelines, but were rarely used and only tracked by necessity	Metrics were regularly collected, but inconsistently used, especially relating to process	Metrics were regularly collected and evaluated/used as intended in process and product domains.	Metrics were properly planned, implemented, proved useful to stakeholders, and were used for process improvement
Comments:				

1. What is the set of metrics that the team is tracking? Has the team gathered these metrics on a consistent basis? What has the team learned from the review of these metrics?

   The team is tracking time spent by all individuals, effort estimation efficiency, requirements volatility, and task slippage. These metrics are gathered on a semi-consistent basis whenever insight is needed into any of these metrics. The team has learned that with a spike in volatility, effort estimation accuracy decreases while task slippage and time spent increases.

Team Effectiveness

Sample Rubric: Team Effectiveness				
Team fractured, unable to work together constructively.	Team had some significant frictions and lack of mutual cooperation.	Team had some significant disconnects and frictions, but worked through them.	Team worked reasonably well together, no problems visible.	Team has achieved synergy, allocated work well, and worked together as a cohesive whole
Comments:				

1. Is this an effective team? What has been contributing to and detracting from the team's effectiveness? What are the team's weak points? What are the team's strong points? What changes can the team make for next term that will make it more effective?

   This is a highly effective team. By leveraging each individual's unique perspective, skills, and expertise, the team is able to implement vast amounts of code and functionality, maintaining the principle that expedited delivery should never compromise code quality. Architectural concerns, documentation issues, and scope and change management are all taken into account before code changes. This ensures that thrashing and pivots are kept to an absolute minimum and all work is productive work. The team's weak point is mostly in tracking metrics, historically it's had trouble with this. Additionally, the team is inexperienced in usability. These weak points either have been or will be fixed by having specific team members in charge of tracking metrics, as well as running comprehensive usability tests with manual and automated testers. Additionally, the team will be recruiting volunteer users to test the usability of the system from the users point of view, rather than functionally. The team is strong at design, implementation, and planning.

Comments: Sponsor can bring board members on for usability testing as administrators and trail managers.

Product

Although Process and Project are important, the product is ultimately why projects are initiated and followed through to completion. An effective Software Engineer maintains a steady focus on the product, and understands the scope of what it entails. Although the list below is not exhaustive, it includes some of the primary areas of importance, especially for the second part of the project.

Design

Sample Rubric: Requirements				
Hardly any design done or documented.	Design significantly inappropriate and missed several key concerns.	Design somewhat inadequate – some concerns not addressed or documented adequately	Design adequate and appropriate and documented. Major design concerns addressed.	Several alternatives explored, eventual design well-suited to application needs.
Comments:				

1. Products need to be designed within guidelines and constraints appropriate for each project. It is also important to consider the impacts of the products that are designed. In the following categories discuss the constraints and impacts that have a bearing on your project. Note that

there may be one or two categories that have no bearing on your project but your project is probably affected by almost all of these.

a. Economic issues

The team is constrained by the resources of the sponsor organization, it cannot use all of the features available to them as they cost money. This impacts how fast certain milestones can be met, and often incurs a cost in time.

b. Environmental issues

The team must take outages into account caused by unstable environmental conditions. This is a constraint of connectivity that is felt by both hardware and software. The hardware must be easy to provision, block, and test as a result.

c. Social issues

As this software is going to be going into public areas, it's important the team take its target audience into account. Hikers and trail managers will be using this platform, and that audience has specific social wants and needs the team must take into account when designing the look and feel of the platform.

- Sponsor: #1 audience is Stewardships Managers, **then** hikers (not 50/50, 90/10).

d. Political issues

It is hoped that the software will be used and managed by the NYS DEC to make political decisions that impact the park, partly on data that they take from this platform. This impacts how the platform must be designed, maintained, and implemented. Additionally, it also imposes the constraint of maintaining correctness, even under adverse conditions.

e. Ethical issues

This platform may be used to drive political decision making. It is the platform's ethical responsibility to provide accurate data without bias.

f. Health and safety

N/A

g. Manufacturability

The manufacturability of devices is constrained by how easy it is to provision them through the website. This process must be quick and easy enough to provision many devices at cost, but not so easy as to be insecure.

h. Sustainability

The platform needs to be operational for many years into the future to be accurate and useful. Thus, the groundwork of the platform must be solid and well engineered to work for many years into the future with minimal refactoring and overhauls.

Implementation

Sample Rubric: Requirements

Implementation quality completely unacceptable. Documentation poor	Implementation artifacts had significant problems or difficult to understand; Implementation is not documented well.	Implementation is marginal or not well thought through. Implementation is sufficiently documented but is hard to understand.	Implementation artifacts reasonably sound. Implementation is well documented and easy to understand.	Implementation artifacts of excellent quality Implementation is well documented and easy to understand
Comments:				

1. How has the development and implementation progressed? What percentage of the product do you estimate is complete at this point? Is the team providing the documentation within the implementation artifacts?

Development and implementation have progressed speedily. By 6/30/26, MVP candidate should be completed with end-to-end functionality from device to user, as well as some optional enhancements. The team provides documentation for all implementation artifacts. As feature freeze is July 1st, no new features will be completed, and instead refinements, documentation, testing, and robustness take priority.

Sponsor: MVP not done until sponsor approves it.

Testing

Sample Rubric: Requirements				
Hardly any testing done. An afterthought.	Testing was completely ad hoc and inadequate.	Test strategy inadequate or inappropriate, testing not thorough.	Testing was reasonably complete, strategy acceptable. Testing has been a major focus of the project.	Very sound strategy, thorough testing of all deliverables, including elements of test driven development.
Comments:				

1. What is the team's testing strategy? Has the team developed a test plan? Is the team performing unit testing? Is the team using any test frameworks, such as JUnit? What are the

testing results to date? Were any major defects found during system test?

The team has a comprehensive testing strategy. There are multiple environments, each with a different purpose. There is the local development environment, mostly unique to each developer, wherein they will checkout a feature branch and implement their code. Once that code is written, it is submitted via PR to the team for review. If any changes are requested, the code is fixed by the developer and resubmitted. In addition to manual reviews, the code is run against the team's suite of integration and component tests.After two approvals the code is merged into TST. Then, the team waits until UAT merge day when all (approved?) changes are merged into TST before performing manual acceptance tests. These manual tests make sure that functionality works as intended from a user's perspective. Then, the new functionality for the sprint is merged to UAT.

In the future, there will be another environment "PROD" in which features that have made their way through UAT are approved by the sponsor and pushed to the official release branch for the project. At this time it is available for public usage. Also in plans are device connectivity and load tests via a device emulator tool. This tool will ensure that all released versions of the devices can interface with the new version of production, and ensures that no breaks in connectivity occur. Additionally, usability testing is planned to ensure that new and advanced users experience a smooth and intuitive look and feel, and that the platform is easy to navigate and use.

Sponsor: PROD isn't really PROD until there is a device network set up. Should instead of being "PROD" should really be "DEMO"

Quality: Verification and Validation

Sample Rubric: Requirements				
Key quality concerns not identified or addressed.	Patchy attention to quality concerns, some concerns missed.	Quality concerns addressed somewhat inadequately.	Reasonably good job of identifying and addressing quality concerns.	Quality concerns explicitly identified and considered fully at every stage.
Comments:				

Quality Improvement Areas

Comments:

Future Considerations:

Final evaluation will weigh heavily on product quality, especially as it relates to customer satisfaction. What are the implications of these at the present point in the project?

Should things go according to the current project plan and the MVP is completed by the end of the current sprint (7/1/2026), the customer should be very satisfied with the server-side component of the product. The customer's previous emails indicate such satisfaction, subject to a few clarifications and minor fixes/enhancements. Following the completion of the other efforts outlined above (e.g. usability testing), the final evaluation should go smoothly.

Sponsor: Clarified what feature freeze entailed.